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# Artificial Intelligence–Powered (Finance) Scholarship†

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*This paper describes a process for generating academic papers using large language models (LLMs) and demonstrates this process's efficacy by producing hundreds of complete papers on stock return predictability, a topic well-suited for our illustration. After mining over 30,000 potential return predictors from accounting data, we generate template reports for 95 signals passing rigorous criteria from the Novy-Marx and Velikov (2024) Assaying Anomalies protocol. These templates detail signal performance predicting returns using a wide array of tests and benchmark performance against more than 200 documented anomalies. Finally, for each template we use state-of-the-art LLMs to generate multiple complete versions of academic papers with distinct theoretical justifications for the observed return predictability, incorporating citations to literature supporting their respective claims. This experiment illustrates the potential of artificial intelligence (AI) for enhancing financial research efficiency, but also serves as a cautionary tale, illustrating how it can be abused to industrialize hypothesizing after results are known (HARKing).* (*JEL* C12, C45, G12, G17)

## 1. *Introduction*

Consider this scenario: A junior professor submits a paper documenting a novel return predictor, which includes precisely formulated hypotheses and robust empirical evidence. The paper is well written, the analysis appears correct, and the hypotheses accurately predict the patterns observed in the data. Should it matter if an AI system generated these

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hypotheses after seeing the results? This question cuts to the heart of how we understand scientific discovery and hypothesis formation, and how our views are being tested by the introduction of LLMs.

In modern academia we face an inherent tension in our treatment of hypothesis formation. We often view post hoc theorizing with suspicion, labeling it as "HARKing" (Kerr 1998). The prevailing academic standard insists that researchers should first develop their theories and predictions and then test them against data. Few significant scientific discoveries in history have, however, adhered to this standard. Legend has it that in 1666 Isaac Newton observed an apple falling from a tree. This observation led him to hypothesize about universal gravitation, forming testable predictions that explained not just falling apples, but planetary motion, tides, and countless other phenomena. Newton developed his hypotheses after observing the phenomena they would later explain. Post-observation hypothesis generation has always been, and will always be, a crucial part of scientific discovery.

This tension between an idealized scientific method and practical discovery is particularly evident in empirical asset pricing, where recent work by Chen, Lopez-Lira, and Zimmermann (2025) shows striking parallels between data mining and traditional peer review. These authors create a large sample of unmotivated data-mined strategies. They then match strategies from this set to published strategies on the basis of their in-sample power predicting cross-sectional differences in average returns. On average these data-mined strategies retain about as much power predicting out-of-sample differences in average returns as the published strategies to which they are matched. They conclude that peer review systematically mislabels mis-pricing or luck as risk, consistent with a view that the anomaly literature often develops ex post theoretical explanations to fit observed empirical patterns rather than testing prespecified economic mechanisms. In fact, the peer review process strongly encourages this. Reviewers and editors often require papers documenting interesting new aspects of the data to include an "economic story," despite the fact that other authors may be better suited to explain the economics behind the results documented in a paper.

The emergence of powerful LLMs has transformed this tension from a matter of scientific practice to one of technological capability. Wiles, Munyikwa, and Horton (2025) show that algorithmic writing assistance can improve job seekers' hiring outcomes without obscuring their actual qualifications, while Horton (2023) explores how LLMs can function as *Homo silicus*—computational analogues of *Homo economicus*. These advances suggest that AI systems can meaningfully engage with economic reasoning and prediction. Building on this, Manning, Zhu, and Horton (2024) present methods for automatically generating and testing scientific hypotheses in silico, though they note that LLMs perform better at qualitative reasoning than quantitative precision. Si, Yang, and Hashimoto (2024) show that LLMs can generate novel research ideas while maintaining high standards of feasibility and scientific merit. Bail (2024), Korinek (2023), and Liang et al. (2024) document LLMs' expanding capabilities across research domains. Most notably, Lu et al. (2024) develop an AI Scientist system that can independently generate research ideas, conduct experiments, and produce papers claimed to exceed typical peer review acceptance thresholds.

Drawing on these advancing capabilities, we demonstrate a complete pipeline for automating academic research production in finance, from hypothesis generation through full paper creation, and use this process to produce hundreds of complete papers. These papers have several distinctive features. They are the outcome of industrial-scale data mining, the hypotheses are ex post, the hypotheses are AI generated, and the papers are AI written.

Using stock return predictability as a testing ground, we first mine accounting data to identify over 30,000 potential predictors.[1] We subsequently apply the Novy-Marx and Velikov (2024) Assaying Anomalies protocol to identify 95 signals that pass rigorous statistical criteria. We then use the accompanying code provided by Novy-Marx and Velikov (2024) to implement their protocol and generate template reports for these 95 signals, detailing signal performance for return prediction using a wide array of tests and comparing the performance of each signal against more than 200 signals documented by published papers in the academic literature.[2]

We then use state-of-the-art LLMs to expand the 95 template reports generated by the Assaying Anomalies protocol into hundreds of complete academic-style papers. We first use Claude Opus 4.1, the most recently released reasoning model by Anthropic, as of the time of this writing, to systematically generate descriptive names for these empirically discovered return predictors, ensuring consistent and meaningful terminology across papers. We then use the same model, employing the template reports as task examples for in-context learning, to machine generate complete academic papers. For each signal, we create four distinct versions of full papers, including abstract, introduction, data, results, and conclusion sections.

The four different paper versions for each return predictor contain distinct hypotheses and economic "explanations" for the same phenomena while each maintains consistency with the empirical findings. We produce the first version using a relatively undirected prompt, which simply asks the LLM to "present economic mechanisms linking the signal to returns" and to "draw on established theoretical frameworks." Because the output of the LLM is probabilistic, not deterministic, we could use these same prompt to generate multiple versions of each paper. To ensure greater dispersion across the different versions associated with each signal, however, we use directed prompts to generated three alternate versions. In particular, we explicitly prompt the LLM to produce one version that develops behavioral finance explanations focusing on slow diffusion of information, one that constructs rational production-based asset pricing arguments, and one that develops rational consumption-based asset pricing theories.

Each of these 380 fully programmatically generated papers contain introductions that follow standard academic conventions, developing theoretical arguments that connect the documented return patterns to established economic mechanisms, incorporating citations to existing (and on occasion, at least for now, hallucinated) literature. Each paper includes comprehensive descriptions of the data and methodology, detailed discussion of results, and contextualized conclusions. All of these papers are available at https://github.com/velikov-mihail/AI-Powered-Scholarship. While the papers and their theoretical frameworks are automatically generated, it is important to note that all empirical analyses and statistical validations are conducted using rigorous methods developed in the academic literature, ensuring the reliability (if not the interpretation) of the underlying findings.

While empirical asset pricing provides an ideal laboratory for demonstrating these issues, the implications extend to any field where researchers develop theoretical frameworks to

[1] This step does not employ LLMs, but has only become feasible recently. Exponential growth in computing power and machine-readable data makes it easy to perform massive data-mining exercises that would have been completely impracticable 20 years ago. While this itself presents both enormous opportunities and enormous challenges for researchers.

[2] While stock return predictability is a topic particularly well suited for this illustration, our results apply to any domain with large accessible datasets, established statistical methodologies, and clear outcome variables, features that encompass much of empirical economics. Other examples include the determinants of capital structure in corporate finance, earnings prediction in accounting, and forecasting economic growth in macroeconomics.

explain empirical patterns. On the one hand, the ability to industrialize hypothesis generation at scale poses serious, potentially existential threats to the peer review process and fundamentally undermines conventional metrics for evaluating scholarly contributions. The ease with which AI can generate plausible theoretical frameworks raises questions about what constitutes genuine scientific contribution versus sophisticated pattern matching wrapped in convincing prose. On the other hand, AI systems can democratize research production, accelerate the pace of discovery, and enable exploration of theoretical spaces at unprecedented scale, allowing researchers to test hypotheses and explore connections across bodies of literature more rapidly than previously possible.

Navigating this landscape will require the academic community to develop new frameworks for understanding research contributions. The future demands enhanced validation systems, new evaluation standards that emphasize out-of-sample validation and novel testable predictions, and above all, a principle that authors bear complete responsibility for any work produced with AI assistance. As Lu et al. (2024) demonstrate in their development of the AI Scientist system, these capabilities are rapidly generalizing across scientific domains, suggesting our findings may foreshadow a broader transformation in how theoretical frameworks are developed across the sciences.

The remainder of the paper is organized as follows. Section 2 describes the process we use to automatically generate papers, including how we mine the data and select return predictors, the prompting we use to produce the text, and how the finished papers are assembled and formatted. Section 3 provides evidence of the process efficacy producing hundreds of papers. It describes the basic structure of the resulting papers, shows that the strength of their results are similar to those found in the existing academic literature, and presents statistics on their readability. Section 4 discusses opportunities the emergent technology offers researches, as well as challenges it poses to traditional methods for evaluating their work. Section 5 concludes.

## 2. *Methodology*

This section outlines the procedures used to identify robust cross-sectional return predictors and subsequently generate manuscript-ready research outputs at scale through AI-driven methods.

### 2.1 *Data-Driven Signal Construction and Filtering*

Our methodology for identifying potential return predictors involves industrial-scale data mining, following the broad framework established in recent literature on factor discovery and cross-sectional asset pricing (Yan and Zheng 2017; Hou, Xue, and Zhang 2020; Chen, Lopez-Lira, and Zimmerman 2025). This involves systematic but traditional computational approaches, testing thousands of potential predictors by creating characteristic-sorted portfolios and running factor model regressions.

We begin by assembling a large set of firm-level candidate return predictors, or "signals," using accounting data from the COMPUSTAT database (Standard & Poor's 1950–2024) and market data from the Center for Research in Security Prices (CRSP 1926–2024). To select the variables, we follow Chen, Lopez-Lira, and Zimmerman (2025) and construct two types of signals from COMPUSTAT data: ratio-type signals formed by dividing one accounting variable by another

TABLE 1
DATA-MINING SUMMARY

| Filter | Number of signals | Percent of filtered |
|---|---|---|
| *Panel A. Data-quality and sample coverage criteria* | | |
| Initial set | 31,460 | |
| Exclude redundant signals | 29,315 | |
| Require 30 stocks | 29,230 | |
| Require data until 12/2024 | 24,754 | |
| Require 360 months | 22,263 | 100.0% |
| *Panel B. Cumulative significance criteria* | | |
| $+ \left\lvert t_{\hat{r}_{(\text{decile, name, EW})}} \right\rvert > 1.96$ | 7,565 | 34.0% |
| $+ \left\lvert t_{\hat{r}_{(\text{quintile, name, EW})}} \right\rvert > 1.96$ | 1,288 | 5.8% |
| $+ \left\lvert t_{\hat{r}_{(\text{quintile, NYSE, EW})}} \right\rvert > 1.96$ | 824 | 3.7% |
| $+ \left\lvert t_{\hat{r}_{(\text{quintile, NYSE, VW})}} \right\rvert > 1.96$ | 652 | 2.9% |
| $+ \left\lvert t_{\hat{\alpha}_{(\text{quintile, NYSE, VW})}} \right\rvert > 1.96$ | 225 | 1.0% |
| $+ \left\lvert t_{\text{Assay, Close Span}} \right\rvert > 1.96$ | 95 | 0.4% |

*Notes:* This table summarizes the sequential filtering of more than 31,000 accounting-based candidate signals into a final set of 95 validated return predictors. Panel A reports data-quality and sample coverage filters, and panel B presents cumulative counts of signals passing successive significance tests. Equal-weighted portfolios assign identical weights to all securities within a portfolio, regardless of firm size. Value-weighted portfolios instead assign weights proportional to each security's market capitalization, reflecting its relative size in the market.

$(X/Y)$, and difference-type signals computed as the change in an accounting variable scaled by a lagged denominator $\left(\Delta X/lag(Y)\right)$. We use 242 potential numerator variables and 65 potential denominator variables from standard COMPUSTAT accounting items, and considering both signal types, we generate an initial universe of 31,460 $(= 242 \times 65 \times 2)$ candidate predictors. These signals span a wide array of firm characteristics, ensuring a rich and diverse search space.

We then implement a series of data-quality and sufficiency filters to refine the candidate set. Panel A of Table 1 reports the results. First, we eliminate redundant measures, where multiple combinations of underlying accounting items produce essentially identical metrics, reducing the universe to 29,315 unique signals. Second, we impose a minimum breadth requirement, retaining only those signals available for enough firms that we can form sufficiently diversified portfolios formed by sorting on the signal in each cross-section. In particular, we require that, after sorting, each portfolio contains a minimum of 30 firms. This criterion reduces the set to 29,230 signals. Finally, to enable robust statistical inference, we require that the signal is available over the complete final 30 years of our sample, which winnows this down to 22,263. These 22,263 candidate signals serve as the initial input into our systematic validation stage.

### 2.2 *Statistical Validation and Robustness Tests*

We subject the signals satisfying the data requirements to a series of increasingly stringent validation tests, designed to identify those that yield economically meaningful and statistically reliable patterns in the cross-section of stock returns. Panel B of Table 1 summarizes these tests.

First, we evaluate the predictive power each candidate signal has to generate a significant difference in average returns between the extremes of portfolios signal-sorted into equal-weighted deciles.[3] Among the 22,263 candidate signals, 7,565 (approximately 34 percent) generate statistically significant return spreads at the 5 percent level. This striking result is consistent with Chordia, Goyal, and Saretto (2020) and Chen, Lopez-Lira, and Zimmermann (2025). Many of the accounting ratios we use are correlated with known returns predictors, especially value, profitability, and investment, and these phenomena are particularly strong among the small capitalization stocks that drive the performance of equal-weighted portfolios. When we narrow further, also considering equal-weighted quintile portfolios, only 1,288 of the previous 7,565 signals (5.8 percent of the initial filtered set) remain significant under this additional more restrictive sorting method.

We then implement more stringent portfolio construction criteria by using New York Stock Exchange (NYSE) breakpoints for quintile formation, a procedure that mitigates potential biases associated with firm size and improves cross-sectional comparability.[4] Under this construction, 824 signals (3.7 percent) also produce significant return spreads using equal-weighted portfolios. Using value-weighted portfolios, which further reduces susceptibility to small-firm effects, 652 signals (2.9 percent) remain significant.

To account for known systematic risks, we next adjust each signal's returns using the Fama and French (2018) six-factor model. After controlling for these well-established risk factors, only 225 signals (1.0 percent) remain statistically significant, suggesting genuine incremental predictive power beyond standard factor benchmarks.

Finally, we subject the remaining 225 signals to the Novy-Marx and Velikov (2024) Assaying Anomalies protocol. This state-of-the-art methodology rigorously benchmarks each candidate predictor against the expansive "zoo" of published anomalies from Chen and Zimmermann (2022), providing a transparent and standardized basis for gauging its relative performance. The protocol also automatically produces detailed "template reports" for each signal, which include comprehensive statistical assessments and robustness checks. After filtering out signals that fail to demonstrate robust performance relative to closely related anomalies, only 95 signals (0.4 percent) survive all layers of validation. These high-quality signals and the associated reports output from the Assaying Anomalies protocol form the foundation for generating full-length academic papers at scale. The empirical results, robustness checks, and all tabular and graphical outputs that appear in the generated manuscripts are drawn verbatim from these template reports produced by the protocol. We make no additional statistical

[3] Portfolio sorts provide a nonparametric method for identifying relations between asset characteristics and their expected returns. Significant differences in time series average returns of well-diversified portfolios constructed to have high dispersion in the candidate return predictor suggest that the candidate predictor has power predicting expected returns, but does so without assuming a functional form on the underlying relation. When the underlying portfolios equally weight the returns of the stocks they hold then the observed relations are primarily driven by small capitalization stocks, which far outnumber large capitalization stocks; when the underlying portfolios are value weighted, then the observed relations are driven by the largest stocks, which disproportionately make up the market by capitalization.

[4] Historically most large capitalization stocks traded on the NYSE. This generates both size and exchange biases when sorting on any stock characteristic correlated with firm size; sorting on the characteristic tends to place large capitalization stocks in one extreme portfolio, which then also tilts toward the NYSE, and small capitalization stocks in the other, which then also tilts toward non-NYSE (primarily NASDAQ and AMEX) stocks. The standard technique for mitigating these biases is using NYSE breaks. NYSE breaks set the characteristic breakpoints (i.e., define the portfolio "buckets") using only stocks that trade on the NYSE. This guarantees an equal number of these typically larger NYSE stocks in each portfolio.

modifications to these outputs; they are inserted directly into the generated papers to preserve the exact quantitative content and formatting of the original programmatic reports.

Figure 1 displays a Sankey plot showing the mapping of accounting variables used to construct stock return predictor signals. It shows the complex network of relationships between accounting variables in our universe of signals. The diagram maps the flow from numerator variables on the left side to denominator variables on the right.[5] Orange connections represent difference-type signals, constructed as the difference between a COMPUSTAT variable and its lag, scaled by a lagged COMPUSTAT variable, while blue connections indicate ratio-type signals formed by dividing one COMPUSTAT variable by another. As we can observe in the figure, certain accounting items feature prominently in our signal construction process. Common stock (CSTK) and debt issuance (DLTIS) serve as the most frequent numerator variables, creating numerous potential signals when paired with various denominator variables.

### 2.3 *AI-Driven Paper Generation Pipeline*

The final stage of the methodology leverages state-of-the-art LLMs and automated text-processing scripts to produce fully formed academic manuscripts for each of the 95 validated return predictors. Here we exploit in-context learning, prompting using the "template report" from the Assaying Anomalies protocol as a task example for each signal. All of the text generated by LLMs in this paper uses Anthropic's most recently released reasoning model as of the time of writing, claude-opus-4–1-20250805.

#### 2.3.1 *Systematic Signal Naming and Classification*

We first apply an AI-powered naming system to assign each validated signal a descriptive, academically credible name. This step produces signal identifiers that integrate COMPUSTAT variable names and acronyms into an informative, nongeneric label.[6]It ensures that each signal is readily interpretable and easily distinguishable within the literature, even as we scale the production of papers across dozens of return predictors.

#### 2.3.2 *Content Generation and Structuring*

Next we generate the core textual content of each paper. The introduction, composed of roughly 1,100 words, is subdivided into four sections to ensure a balanced, academically coherent narrative:[7]

(i) Motivation (∼200 words): Frames the research question within the broader asset pricing literature, discussing market efficiency, cross-sectional predictability, and recent developments in factor research.

[5]The chart type is called a "Sankey" after Captain Matthew Henry Phineas Riall Sankey, who popularized width-proportional energy flow diagrams in 1898 to explain steam engine efficiency. In our case, each numerator–denominator pair is drawn as a single, equal-width link. Thus, what conveys importance is the accumulated bundle: Variables with more outgoing (or incoming) links form visibly thicker stacks at their node, making it easy to spot hubs and broad mappings.

[6]Appendix A.1 contains our specific prompts.

[7]Appendix A.2 contains our specific prompts.

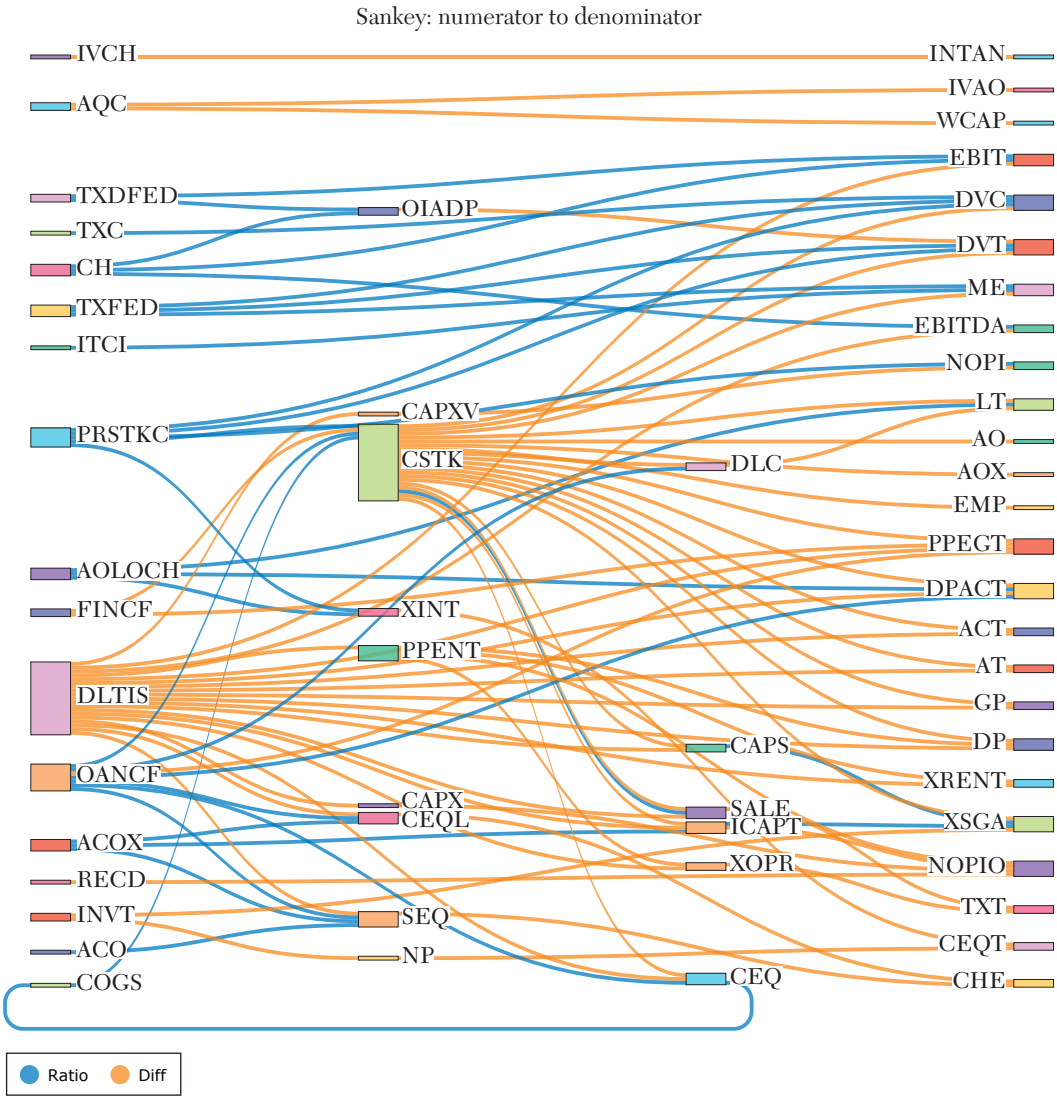



*Figure* 1. Sankey Plot

*Notes:* This figure displays a Sankey plot showing the mapping of accounting variables used to construct stock return predictor signals. The figure illustrates the flow from numerator variables (left) to denominator variables (right). Items in the middle columns appear as both numerator and denominator variables. There is no conceptual distinction between the entries in the two middle columns; they are separated only to improve readability. The line from CEQ to COGS is the one exception where the flow is from right to left. Orange connections represent difference-type signals (constructed as the difference of a COMPUSTAT variable and its lag, scaled by a lagged COMPUSTAT variable), while blue connections represent ratio-type signals (constructed as the ratio of two COMPUSTAT variables). The initialisms in this figure correspond to the items listed in Table B1.

(ii) Hypothesis development (∼300 words): Here we use four different prompts to produce the four different versions of each paper. One is relatively undirected, while three ask for hypotheses from distinct important strands of the literature. The four different prompts here direct the LLM to:
  (a) Propose economic mechanisms justifying the signal's predictive power, citing relevant theoretical and empirical studies to maintain a scholarly tone and contextualize the new factor;
  (b) Propose behavioral finance explanations focusing on limited attention and slow diffusion of information, linking the signal to gradual information incorporation and underreaction, citing seminal work on investor inattention and news dispersion;
  (c) Propose rational production-based asset pricing arguments by mapping the signal to marginal costs, investment frictions, or conditional risk premia in a production economy, discussing how the signal proxies for exposure to shocks to productivity, adjustment costs, or investment opportunities, and relating cross-sectional pricing patterns to production-based risk factors;
  (d) Propose rational consumption-based asset pricing theories that tie the signal to exposure to consumption risk or marginal utility variations (e.g., long-run risks, habit formation), discuss state dependence across good versus bad economic times and sensitivity to disaster or macroeconomic shocks, and relate findings to consumption-based factors and the intertemporal marginal rate of substitution.

(iii) Results summary (∼300 words): Presents key empirical findings, highlighting statistical significance, robustness checks, and comparisons to established anomalies.

(iv) Contribution (∼300 words): Places the proposed signal in relation to three–four closely related studies, articulating how the new evidence enhances our understanding of systematic return drivers and contributes to ongoing debates in the literature.

Our prompts specifically ask for all generated text to adhere to a formal academic writing style and to use an active voice. We also ask that the text carefully distinguishes correlation from causation, avoids unwarranted claims, and ensures appropriate application of tense to reflect established knowledge versus new findings. Citations are embedded using LaTeX-formatted references, and all writing conventions align with norms in leading finance journals.

The other added sections of each manuscript, including data and conclusion sections, are generated following similarly structured prompts.[8]

### 2.3.3 *Document Assembly, Formatting, and Quality Assurance*

The final assembly relies on custom scripts to incorporate the generated content into a standardized LaTeX template. We maintain a consistent document structure by programmatically inserting AI-generated sections into the appropriate manuscript components, preserving academic formatting standards and internal consistency.

We maintain dedicated, signal-specific .bib files for references, merging newly introduced citations into a base bibliography. This step ensures that each manuscript is properly referenced and that the supporting literature is consistently integrated into the text.

[8] Appendices A.3 and A.4 contain our specific prompts for the data and conclusion sections, respectively.

The final documents are compiled using a multi-pass LaTeX build process to ensure proper formatting, stable referencing, and a professional appearance. Automated cleaning procedures remove extraneous auxiliary files and streamline file management. The end product is a fully formed, academically styled PDF suitable for journal submission. This fully integrated pipeline, encompassing data-driven predictor identification, statistical validation, AI-based content creation, and automated document preparation, demonstrates a scalable approach to generating and disseminating academic finance research papers. All the code used to construct this pipeline is available at https://github.com/velikov-mihail/AI-Powered-Scholarship.

## 3. *Results*

### 3.1 *Scale and Efficiency of AI-Powered Research*

The automated pipeline successfully generates complete academic papers. It documents 95 novel return predictors, writing four distinct papers for each that provide different economic motivation for the observed phenomena.

Figure 2 illustrates the complete structure and content of a representative AI-generated paper from our corpus of 380 manuscripts, specifically showcasing what the LLM designates as the liquidity leverage intensity (LLI) signal.[9] Figure 3 provides a thumbnail grid of the complete AI-generated paper produced under the slow diffusion of information (SDI) hypothesis for the same signal.[10] The exhibits demonstrate the comprehensive nature of our automated paper generation pipeline, displaying a fully formed academic manuscript that spans 30 pages and includes 8 detailed figures and 5 tables. Each generated paper follows a standardized yet academically rigorous format, beginning with a substantive introduction that develops economic hypotheses linking the signal to expected returns, followed by systematic empirical sections documenting signal construction, diagnostic tests, portfolio sorts, factor model alphas, and comparisons with the broader anomaly zoo. This standardized structure ensures that despite the industrial scale of production, each paper maintains at the very least the appearance of depth and rigor expected of academic finance research, complete with robustness tests, spanning regressions with related anomalies, and conditional sorts that are expected in this type of finance paper. The consistency and professional presentation of these AI-generated papers, as exemplified in these figures, underscores both the potential and the challenge our experiment poses for the academic review process and for research more generally.

Table 3 in Appendix B documents all of the signals and the resulting paper names. The process is remarkably efficient. While the data mining, validation, and template report generation from the Assaying Anomalies protocol takes about a day of computation time, the final paper generation takes about 12 hours. This represents a dramatic acceleration compared to traditional research paper development.

[9] Our code organizes the accounting ratios alphabetically, by COMPUSTAT numerator and then COMPUSTAT denominator. While this organization is arbitrary, we tend to use ACO to SEQ in our concrete examples, which the LLM named "liquidity leverage intensity" because it is at the top of our list of 95 candidate predictors. While this exemplar signal choice is essentially arbitrary, it is fairly representative in the sense that it yields a strategy Sharpe ratio near the average of those employed in our automatic paper production pipeline.

[10] The full "template report" for the LLI signal is reproduced in Supplemental Appendix A, while the full paper shown in Figure 3 is reproduced in Supplemental Appendix B.

**Paper Title:** "Liquidity Leverage Intensity and the Cross Section of Stock Returns"
**Author:** I. M. Harking

**Abstract:**
This paper studies the asset pricing implications of Liquidity Leverage Intensity (LLI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2024). A value-weighted long/short trading strategy based on LLI achieves an annualized gross (net) Sharpe ratio of 0.38 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 21 (20) bps/month with a *t*-statistic of 2.79 (2.76), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration) is 17 bps/month with a *t*-statistic of 2.37.

**Document Structure:**

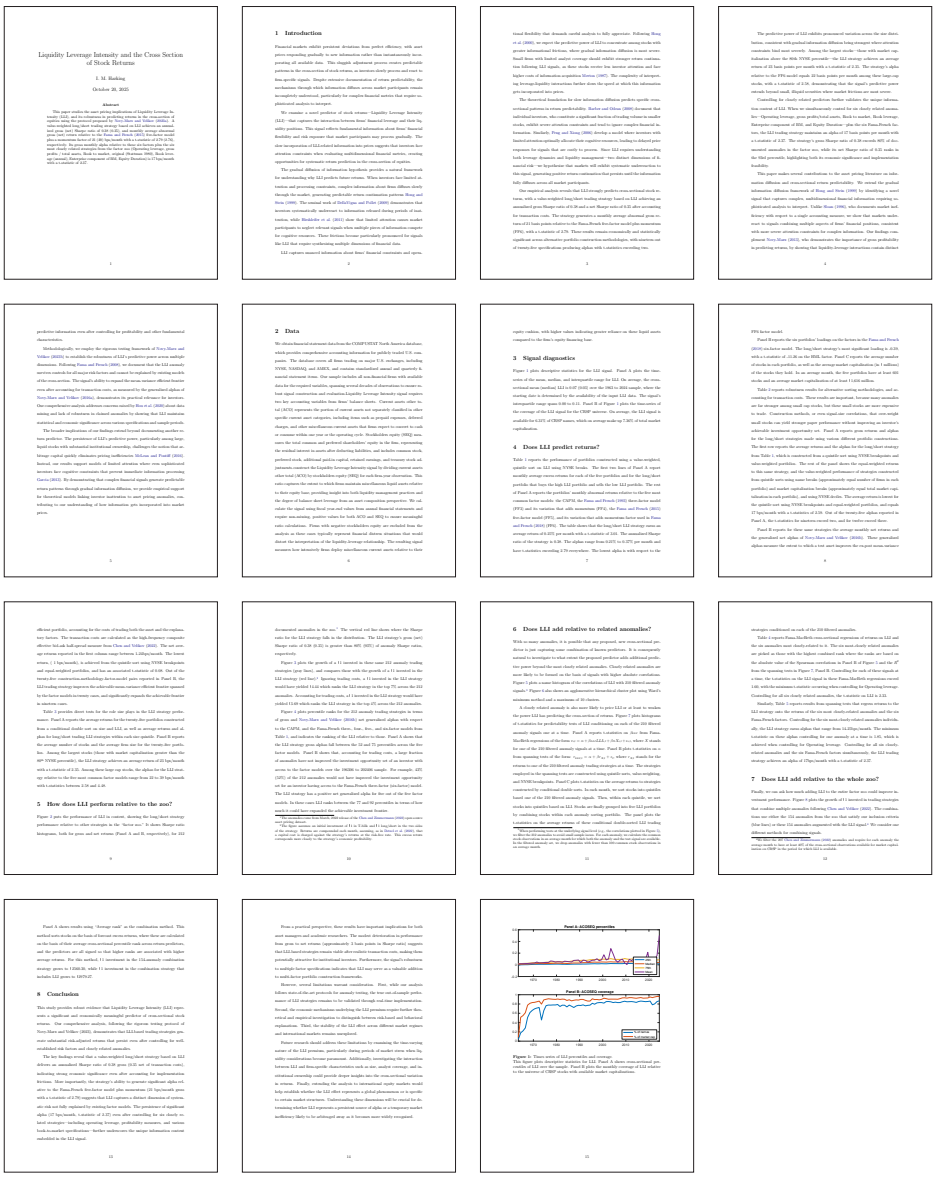
| Section | Pages | Content description |
|---|---|---|
| 0. Title Page | 1 | Title, abstract |
| 1. Introduction | 4 | Economic motivation and hypothesis development |
| 2. Data | 1 | Signal construction using COMPUSTAT variables |
| 3. Signal Diagnostics | 1 | Time-series properties and coverage statistics |
| 4. Return Prediction Tests | 2 | Portfolio sorts and factor model alphas |
| 5. Comparison to Factor Zoo | 2 | Sharpe ratio and alpha percentile rankings |
| 6. Related Anomalies Analysis | 1 | Correlation and spanning tests with similar signals |
| 7. Combination Strategies | 1 | Incremental value in multi-signal portfolios |
| 8. Conclusion | 2 | Summary and implications |
| Tables and Figures | 13 | Results from "Assaying Anomalies" protocol |
| References | 2 | Academic citations |

**Tables and Figures:**

| Tables | Figures |
|---|---|
| Table 1: Basic sort results (returns, alphas, factor loadings) | Figure 1: Time series of LLI percentiles and coverage |
| Table 2: Robustness to sorting methodology and trading costs | Figure 2: Distribution of Sharpe ratios (comparison to 212 anomalies) |
| Table 3: Conditional sort on size and LLI | Figure 3: Dollar invested growth charts |
| Table 4: Fama–MacBeth regressions with related anomalies | Figure 4: Gross and net alpha percentiles |
| Table 5: Spanning tests with related anomalies | Figure 5: Distribution of correlations with other signals |
| | Figure 6: Hierarchical cluster plot |
| | Figure 7: Distribution of t-stats on conditioning strategies |
| | Figure 8: Combination strategy performance |

*Figure* 2. Sample AI-Generated Paper Content and Structure

*Notes:* This exhibit displays the complete abstract from an AI-generated paper produced under the SDI hypothesis for the LLI signal, along with the document's structure and comprehensive empirical content. Each generated paper follows this standardized format, containing approximately 30 pages of analysis including eight figures and five tables documenting the signal's return predictability and robustness.




(continued)

*Figure* 3. Thumbnail Grid of the PDF Document

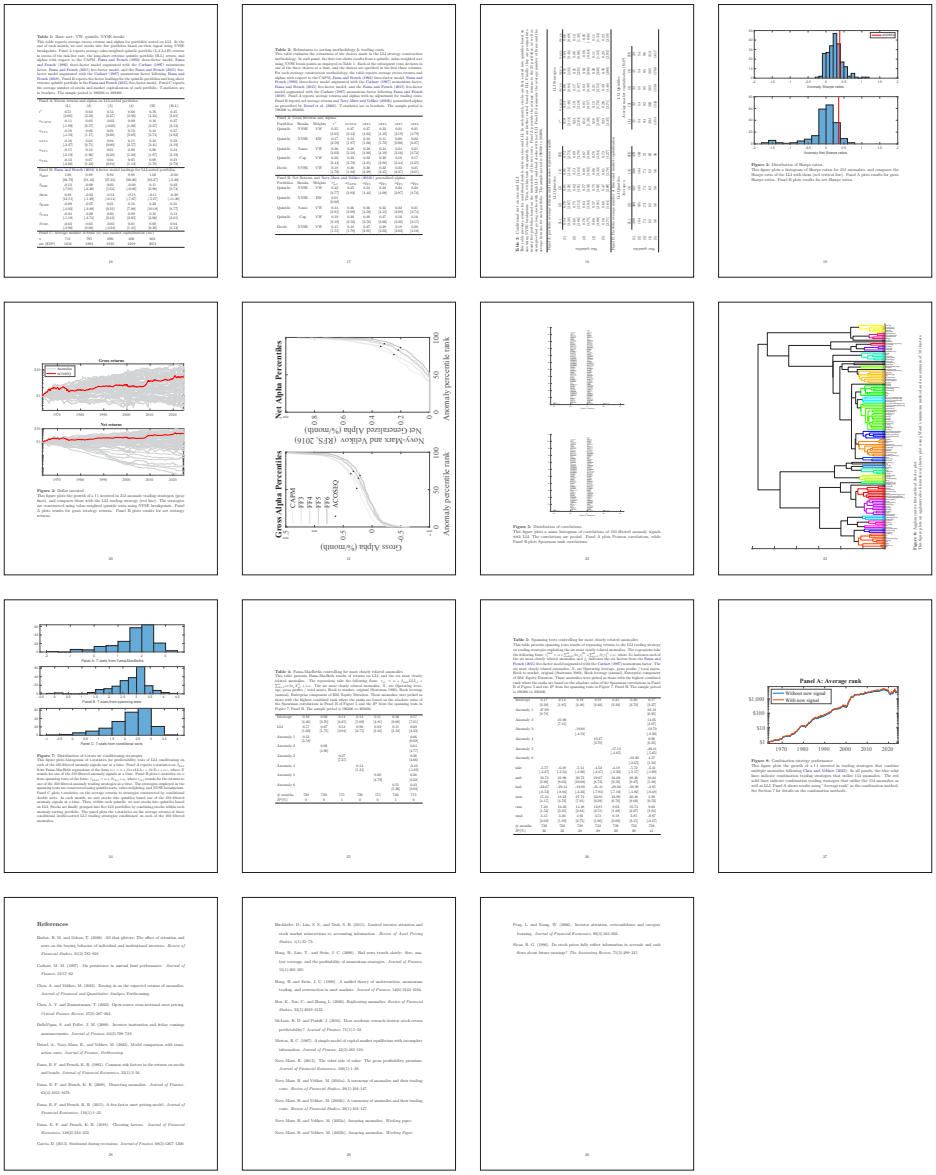


*Figure* 3. Thumbnail Grid of the PDF Document (continued)

*Notes:* This figure presents a two-page thumbnail grid of the complete AI-generated paper produced under the SDI hypothesis for the LLI signal. Each panel displays a matrix of miniature page images arranged in a 4×4 grid (16 pages per panel), collectively showing all 30 pages of the full manuscript across the two panels.

### 3.2 *Quality of Generated Content*

The AI-generated papers exhibit several notable characteristics that make them superficially difficult to distinguish from human-authored research. The signal-naming algorithm produces descriptive and economically meaningful names that avoid generic terminology. For example, the ratio of other current assets (ACO) to shareholder equity (SEQ) becomes "liquidity leverage intensity," while the difference in acquisitions (AQC) to working capital (WCAP) is termed "acquisition capacity utilization." The generated introductions are written in academic prose, consistently identifying plausible economic mechanisms, integrating findings with existing literature through appropriate (though occasionally hallucinated) citations, and developing testable hypotheses that align with the empirical results. Each paper positions its contribution within the broader literature, citing papers from top finance journals and highlighting methodological innovations. The data sections provide technically accurate descriptions of signal construction, successfully translating COMPUSTAT variable codes into meaningful economic quantities while maintaining precise documentation. The conclusions synthesize findings without appearing formulaic, paraphrasing the abstract while adding forward-looking research suggestions.

To evaluate the quality of the data-mined signals, Figure 4 shows the distribution of $t$-statistics on the average excess returns earned by trading strategies. It shows that the distribution for our data-mined signals closely mirrors those from the 212 signals in the published anomaly literature documented in Chen and Zimmermann's (2022) Open Source Asset Pricing (OSAP) database. The equal-weighted comparison (top panel) is striking. The red distribution of data-mined signals overlaps almost perfectly with the blue distribution of published anomalies from OSAP, 87 percent of which are equally weighted, suggesting that our criteria for selecting the strongest mined signals is similarly strenuous to that historically provided by the peer review process. It also suggests, consistent with the findings in Chen, Lopez-Lira, and Zimmermann (2025), that much of the published anomaly literature may be statistically indistinguishable from sophisticated data-mining exercises. The value-weighted comparison (bottom panel) shows more differentiation. The value-weighted performance of the mined strategies, which do not disproportionately overweight the microcaps stocks driving many anomalies, exhibits a shorter right tail of extremely significant results.

Figure 5 attempts to evaluate the quality of the LLM-generated content by comparing the readability of our AI-generated introductions to 140 published papers in the OSAP database using two standard metrics: the Flesch–Kincaid index (FKI) and the Gunning fog index (GFI). The FKI measures text complexity based on average sentence length and syllables per word:

$$\text{FKI} = 0.39 \times \left(\frac{\text{words}}{\text{sentences}}\right) + 11.8 \times \left(\frac{\text{syllables}}{\text{words}}\right) - 15.59.$$

The GFI focuses on the percentage of complex words (three or more syllables) and sentence length:

$$\text{GFI} = 0.4 \times \left[\left(\frac{\text{words}}{\text{sentences}}\right) + 100 \times \left(\frac{\text{complex words}}{\text{words}}\right)\right].$$

Both metrics are meant to estimate the years of formal education required to comprehend a text. The box plots reveal that AI-generated introductions cluster tightly around readability scores of 16–18 years of education required (roughly college graduate), with remarkably low

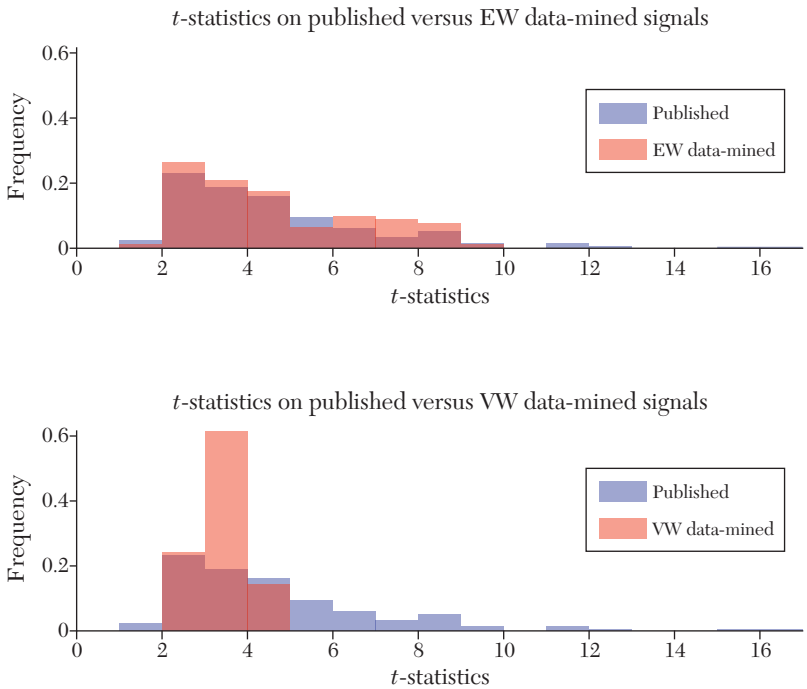



*Figure* 4. Comparison of Published versus Data-Mined Signals

*Notes:* This figure displays overlapping histograms of $t$-statistics on the average excess returns earned by trading strategies based on published versus data-mined return predictors. The "Published" distribution is based on 212 $t$-statistics taken directly from 142 published anomaly papers and documented in Chen and Zimmermann's (2022) *Open Source Cross-Sectional Asset Pricing* library, while the "data-mined" distributions is based on 95 trading strategies based on signals discovered using our automated search pipeline. The data-mined trading strategies are constructed using quintile sorts with NYSE breakpoints and equally weighted (EW, top panel) or value-weighted (VW, bottom panel) portfolios. Each histogram is probability normalized, plotting $t$ for published (blue) and data-mined (red) signals.

variance across all four prompt variations: SDI, production-based (prod), consumption-based (cons), and unrestricted (free). Published papers show greater dispersion in readability, with median scores slightly lower (around 13–16 years) but with notable outliers spanning from high school to postdoctoral complexity levels.[11] Although partially mechanical due to the structured prompt, the AI's consistency is double edged: While it reliably produces appropriately sophisticated academic prose, the narrow distribution suggests a formulaic quality that lacks the stylistic diversity of human academic writing. The different theoretical prompts produce nearly identical readability profiles, indicating that the LLM consistently adheres to the requested structure regardless of the economic framework it develops.

[11]These results are broadly consistent with Walther and Dutordoir (2025), who document an increase in the use of generative AI for writing in academic finance.

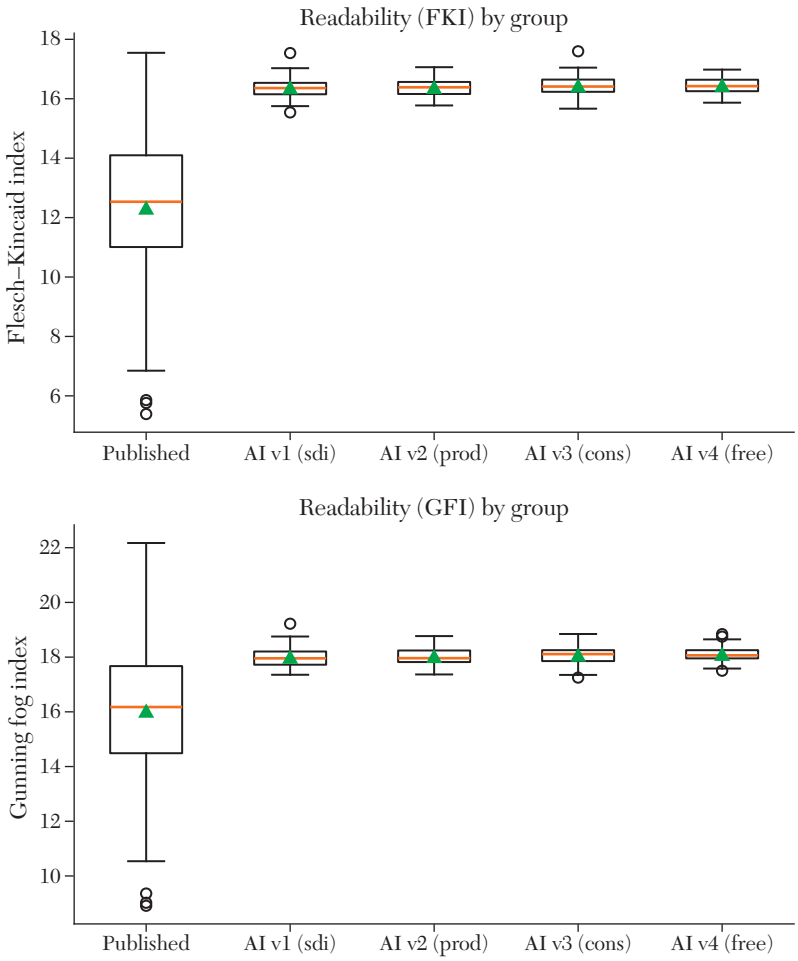



*Figure* 5. Box Plots of Intro Readability

*Notes:* This figure compares the readability of introductions from published finance papers with four versions of AI-generated papers using two standard readability metrics. The top panel shows FKI scores and the bottom panel shows GFI scores, both measuring the years of education required to comprehend the text. The box plots display distributions for published papers from top finance journals and four AI-generated versions: v1 (SDI) using slow diffusion of information prompts, v2 (prod) using production-based asset pricing prompts, v3 (cons) using consumption-based asset pricing prompts, and v4 (free) using unrestricted theoretical development. Each box plot shows the median (horizontal line), interquartile range (box), whiskers extending up to 1.5 times the interquartile range from the first and third quartiles, and outliers (circles).

Ultimately, determining the true quality of these AI-generated papers requires a field experiment, submitting them to peer review alongside human-authored research to see if referees can distinguish between them and whether they pass the profession's quality standards. Without such validation, we cannot definitively assess whether these signals meet the

conventions of scientific credibility or are simply sophisticated nonsense wrapped in convincing prose. It may well be that some of these 95 signals capture real economic phenomena that have been overlooked by researchers, making them legitimate contributions to the literature despite their automated origins.

## 4. *Challenges and Opportunities*

The inherent difficulties in determining AI-powered research quality is, in fact, central to the point of our paper: If we cannot easily determine the true validity of the claims in the generated papers even knowing their provenance, how much harder will this task become when similar AI-generated research enters the publication pipeline without disclosure? The inability to distinguish between mechanical pattern recognition with post hoc theorizing and genuine economic insight poses an existential challenge to our traditional peer review process and our understanding of what constitutes genuine scientific contribution.

### 4.1 *Challenges to the Existing Paradigm*

The scale and automation of AI-powered research generation raises fundamental concerns about the integrity of knowledge production in the academic community. The profession has institutional safeguards in place against potential abuse of data mining and post hoc theorizing. Perhaps most importantly, the profession rewards (at least in the long run) scholarly reputation built through sustained contribution of influential work rather than mere quantity of publications. Researchers who consistently publish low-quality papers, even in substantial numbers, rarely achieve a field's highest honors or most prestigious appointments. The peer review process provides additional screening, with referees and editors scrutinizing not just statistical significance but theoretical foundations, methodological rigor, and broader contribution to the literature (though, as noted earlier, for good or for bad this process in practice probably encourages more HARKing). The practice of presenting work at research seminars and conferences before publication creates opportunities for detailed questioning about theoretical mechanisms and research design choices. The recent increasing emphasis on replicability, including requirements for publicly sharing data and code, has added another layer of quality control. In a world powered by AI, this requirement is especially important because the same tools that make testing easier, and thus raise overfitting concerns, also lower the bar on replication and independently verifying robustness through additional tests.

These various institutional safeguards have generally served the profession well in maintaining research standards. The emergence of sophisticated AI systems capable of generating (multiple) plausible theoretical frameworks at scale, however, poses novel challenges to these traditional mechanisms. Chen and Dim (2025) show that rigorous large-scale data mining of accounting data produces predictive signals with strength comparable to the anomalies published in top finance journals. When AI systems can rapidly produce hundreds of seemingly coherent theoretical explanations for mined empirical results, how do we maintain meaningful quality control through our existing institutions and avoid overwhelming our traditional peer review process? The peer review process has already been strained by rapid growth in our profession, which has far outstripped the growth in the number of high-quality outlets for publishing our work; a dramatic increase in

the rate of paper production could break it. If we were to submit the 380 papers produced for this project to peer reviewed journals, the opportunity cost to the profession of reviewing the work would easily run to hundreds of thousands of dollars. If even a small percentage of the profession settled on this sort of strategy it would completely swamp the journals.

Our work also suggests challenges facing academia, particularly in how it hires and promotes its members. Each of our AI-generated papers naturally includes citations to existing literature to support its hypothesis development. When scaled to hundreds or thousands of papers, this automated citation generation could artificially inflate citation counts, including our own.[12] The ease with which AI can generate convincing theoretical frameworks that reference prior literature may inadvertently create a new form of academic arbitrage, where researchers can boost their citation counts through automated paper generation.[13] It is easy to imagine scenarios in which entire fictitious subfields of a literature emerge in which all of the citations are from AI-generated papers to other reciprocally citing AI-generated papers. It is hard to imagine a task more suited to LLMs' current capability than perpetrating a large-scale version of the "Sokal hoax."[14] The debasement of the informativeness of citation metrics widely used in high stakes decisions by promotions and tenure committees presents profound challenges to all of academia.[15]

### 4.2 *Opportunities for Future Researchers*

While our demonstration of AI-powered academic paper generation raises important questions about research integrity, validation, and the role of theory in empirical finance, it also demonstrates the potential gains from automating aspects of research production. First, our results show that AI can now develop hypotheses at an unprecedented scale. This capability fundamentally changes how we might approach the relation between empirical findings and hypothesis development. While traditional research typically starts with hypothesis development followed by empirical testing, AI enables rapid iteration between empirical discovery and theoretical justification. Second, this automated approach could democratize research production by reducing barriers to entry. The ability to quickly generate and test multiple hypotheses could accelerate the discovery of market inefficiencies, and might also lead to their

[12] If Google's algorithms pick up the 95 paper titles we've generated and posted on our GitHub page, that would result in at least $95 \times 3 = 285$ (Novy-Marx) and $95 \times 4 = 380$ (Velikov) additional Google Scholar citations for our own work (each paper cites Novy-Marx at least three times and Velikov at least four times; it is entirely possible that our instructions to the AI that generate citations to related literature yield additional citations to our own work). If Google's algorithms counted the four distinct versions of each paper identified with a common title as distinct papers (and we could effortlessly generate distinct titles), then these minimum numbers would be 1,140 and 1,520, respectively.

[13] These concerns are not merely theoretical. The Google Scholar page for Yoesoep Edhie Rachmad (since made unavailable) recently reported more than 3.75 million Google Scholar cites, driven primarily by a few thousand largely AI-generated "papers" (each only a few paragraphs) that primarily cite each other.

[14] The "Sokal hoax" refers to physics professor Alan Sokal's (1996) unauthorized tests on human subjects (journal editors) of the hypothesis that "a leading North American journal of cultural studies" would publish any "article liberally salted with nonsense if (a) it sounded good and (b) it flattered the editors' ideological preconceptions." His erudite-sounding gobbledygook, "Transgressing the Boundaries: Toward a Transformative Hermeneutics of Quantum Gravity," was accepted and published in short order.

[15] It is hard to imagine Google's simple citation count remaining relevant. Meaningful metrics must essentially ignore "ghost citations." They will consequently need to weight citations by impact, perhaps by their own and/or that of the outlets in which they appear.

faster elimination through increased trading activity as these findings become more widely disseminated.

AI's impact on research is no longer speculative, it's here. Deep research systems now synthesize hundreds of sources into comprehensive literature reviews in minutes, completing work that once took weeks (Korinek 2025). Lu et al.'s (2024) AI Scientist generates research ideas, runs experiments, and writes complete machine learning papers that meet top-tier conference standards for the cost of a couple of cups of gourmet coffee. The implications are staggering. In mathematics, progress has been even more dramatic. AI systems have progressed from solving geometry problems at an undergraduate level to achieving gold-medal performance at the 2025 International Mathematical Olympiad, with systems from Google DeepMind and OpenAI each independently scoring 35 out of 42 points and solving problems in algebra, number theory, and geometry that challenge elite human mathematicians (Cai and Singh 2025). Google DeepMind's AlphaEvolve system recently improved the best-known solutions to 20 percent of over 50 open problems in mathematical analysis and broke a 50-year-old record in matrix multiplication algorithms (Google DeepMind 2025). These aren't incremental advances in research productivity. AI systems are beginning to discover genuinely novel solutions to hard problems. Current systems still need human oversight and they can't yet formulate entirely new research questions or create new fields of mathematics. But the trajectory is clear, and the pace of improvement suggests we may not have to wait long.

## 5. *Conclusion*

### 5.1 *A Step in an Evolving Landscape*

This study represents an early exploration of AI's capabilities in producing academic finance research at scale. The tools and methods demonstrated here are in their infancy, and we expect significant advances in the coming years. Just as the last decade witnessed breakthroughs in computational finance, we anticipate rapid evolution in AI-powered research tools.

Current limitations, such as occasional citation hallucination, ambiguous theoretical framing, and rote replication of established narratives may become trivial to solve as LLMs become more context aware and evidence based. Future systems will likely incorporate automated fact-checking mechanisms, dynamic citation verification, integrated replication capabilities, and self-improving research validation. As these technologies mature, the boundary between human-generated and AI-assisted research may become increasingly fluid, necessitating new frameworks for understanding and evaluating scholarly contributions.

### 5.2 *Future Directions and Recommendations*

AI represents more than just a technological advancement in the research process; it has the potential to fundamentally shift how we generate and validate knowledge. The ability to automatically generate and test hypotheses will require us to reconsider what even constitutes meaningful academic contribution. This enormous potential, however, comes hand in hand with great challenges. While it is impossible to fully forecast all the possible dangers AI poses

to the process of scientific exploration, we can identify a few areas crucial to develop in order to maintain research integrity in an AI-enabled environment.

First, it is critical that authors are held completely accountable for any mistakes they make, even (or especially) when those works employ AI. It is currently common for institutions of higher learning, news outlets, and academic journals to require full "disclosure and attribution" of AI usage. Such policies are insufficient to ensure full disclosure and may even undermine responsible AI use. AI is powerful, but ultimately just a tool. People do not cite their tool. Generative AI might, while it is novel, seem like a source (or even a coauthor), but as it becomes ever more pervasive and its usage more mundane it will permeate some aspect of everything we do. There is no point to, or information to be gleaned from, a policy that requires everyone to always disclose and attribute work to a tool that everyone is always using. Perhaps more perniciously, the policy can even encourage sloppy AI usage. The attribution requirement currently provides some authors an excuse for the shoddiness of their output, allowing them to deflect deficiencies in their work to the AI that they used. Blaming AI for shortcomings in one's work should be completely unacceptable. If a craftsperson cannot use a tool to produce high-quality work, then it is on the craftsperson to avoid the tool. Requiring disclosure and attribution of AI usage does not encourage responsible AI usage; holding authors fully responsible for any work product produced using AI does.

Another priority should be developing enhanced validation systems and establishing new quality control mechanisms for AI-assisted research. We need automated tools that can verify citations, ensure reference accuracy, and validate theoretical frameworks. These systems should be capable of detecting circular reasoning, redundant theorizing, and hallucinated citations. Future iterations of research automation should incorporate these built-in quality controls, including automated checking of theoretical consistency, detection of overlapping or redundant hypotheses, validation of citation networks, and integration with replication databases. These mechanisms would help ensure that AI-generated content maintains high academic standards while contributing meaningful insights to the literature.

The academic community also needs to develop new evaluation standards that reflect the realities of AI-enabled research production. These standards should place greater emphasis on out-of-sample validation and focus on practical implementation and economic significance. Economic stories justifying observed phenomena should be evaluated, at least in part, by the novel testable predictions they make beyond the primary findings they were reverse engineered to explain.

Implementing these recommendations requires significant coordination within the academic community, or (perhaps more likely) mechanisms that incentivize their adoption. Such efforts are essential, however, for maintaining research integrity as AI capabilities continue to advance. The focus should shift from merely identifying potential problems to developing practical solutions that can be implemented across the field. This includes establishing strict standards regarding author responsibility for AI-generated work product, creating shared databases for validation, and developing community-wide standards for evaluating AI-assisted research.

The questions posed here have no easy answers, but demand careful consideration as we enter an era in which AI becomes an increasingly integral part of the research process. The future of research may depend less on our ability to generate hypotheses and more on our capacity to distinguish meaningful insights from statistically significant but theoretically hollow findings.

## *Appendix A. LLM Prompt Engineering*

This appendix details the exact prompts used to generate the papers' content. Each prompt was designed to ensure consistent, high-quality academic writing while maintaining appropriate structure and theoretical development. The model used for generation was Claude Opus 4.1 (claude-opus-4–1-20250805).

### *A.1 Signal-Naming Prompt*

The signal-naming prompt was designed to generate economically meaningful three-word names with corresponding acronyms. The prompt begins with:

**Signal Naming Prompt—Introduction**

You are naming a financial signal for an academic finance paper. Create a THREE-WORD NAME that conveys specific ECONOMIC INTUITION about what the signal measures.
Signal type: `[signal_type]`

For ratio-type signals, the prompt specifies:

**Ratio Calculation**

Calculation: '`[numer_full]`' divided by '`[denom_full]`'

For difference-type signals:

**Difference Calculation**

Calculation: Change in '`[numer_full]`' scaled by '`[denom_full]`'

For signals involving negative transformations:

**Negative Transformation**

Note: The signal uses the negative of this value.

To ensure uniqueness across multiple signals, the prompt includes a duplicate avoidance mechanism:

**Duplicate Avoidance**

IMPORTANT: The following names are ALREADY USED. You must create a DIFFERENT name:

- `[name_1]`
- `[name_2]`
- …
- (… and `[N]` more)

The prompt provides examples of well-formed names:

**Examples**

EXAMPLES of good three-word names:
'Cash Flow Volatility'
'Asset Growth Rate'
'Operating Profit Margin'
'Working Capital Efficiency'
'Earnings Quality Score'
'Financial Constraint Index'

The prompt then specifies detailed guidelines:

**Naming Guidelines**

GUIDELINES:

1. Must be EXACTLY THREE WORDS
2. Think: What economic story does this ratio/change tell?
   - What firm behavior or characteristic leads to high values?
   - What does this reveal about management decisions or firm quality?
3. Consider the SPECIFIC relationship between numerator and denominator
   - Don't just use generic terms like 'liquidity' or 'flexibility'
   - Think about what makes THIS combination meaningful
4. Use precise finance terminology (e.g., 'margin', 'turnover', 'coverage', 'intensity', 'quality')
5. Avoid: 'ratio', 'difference', 'scaled', 'delta', 'factor', 'metric'
6. Avoid: Generic adjectives like 'strategic', 'dynamic', 'enhanced'
7. Avoid: Negative words–use positive framing even for low-is-good signals
   - Use 'efficiency' not 'inefficiency'
   - Use 'quality' not 'poor quality'
   - Use 'stability' not 'instability'
   - Use 'constraint' not 'lack of constraint'
8. Make each name DISTINCTIVE–no repeating the same concept across signals
9. Acronym: 3–5 uppercase letters derived from the name

Ask yourself: If this signal predicts returns, what economic mechanism explains it? (e.g., profitability? efficiency? growth? risk? mispricing? constraints?)

Finally, the response format is specified as:

**Response Format**

Respond inside `<response></response>` tags with strict JSON:

```
{
  "name": "Three Word Name",
  "acronym": "ACRO"
}
```

The implementation includes a retry mechanism (up to 3 attempts) that adds a message on subsequent attempts to emphasize the need for a different name when duplicates are detected.

### *A.2 Introduction Generation Prompt*

The introduction prompt was enhanced with hypothesis-driven guidance. The system dynamically selects one of three theoretical lenses (slow diffusion, production based, or consumption based) or uses no specific guidance:

**Finance Paper Introduction Prompt**

Write an introduction for a finance academic paper discussing the signal '`[signal_name]`' that predicts stock returns. Please follow these detailed guidelines:

**Hypothesis lens to adopt (be explicit and consistent throughout):**

- Lens: `[hypothesis_label]`
- What to emphasize:
  - `[hypothesis_bullet_1]`
  - `[hypothesis_bullet_2]`
  - `[hypothesis_bullet_3]`

**Structure and Style:**

- 1. Motivation (2 paragraphs, ∼200 words total):
  - Open with a broad statement about market efficiency or asset pricing.
  - Identify the specific gap or puzzle in the literature.
  - Use active voice and declarative statements.
- 2. Hypothesis Development (3 paragraphs, ∼300 words total):
  - Present economic mechanisms linking the signal to returns.
  - Draw on established theoretical frameworks.
  - Build logical arguments step by step.
  - Support each claim with citations to foundational papers in LaTeX format.
- 3. Results Summary (3 paragraphs, ∼300 words total):
  - Lead with the strongest statistical finding.
  - Present results in order of importance.
  - Use precise statistical language.
  - Include economic significance.
  - Mirror exactly the terminology used in the results section.
- 4. Contribution (3 paragraphs, ∼300 words total):
  - Position relative to 3–4 most closely related papers.
  - Cite papers from: *Journal of Finance*, *Journal of Financial Economics*, *Review of Financial Studies*, *Journal of Accounting Research*, *Journal of Accounting and Economics*.
  - Highlight methodological innovations.
  - Emphasize novel findings.
  - End with broader implications.

The hypothesis guidance varies based on the selected theoretical lens.

### A.2.1 *Slow Diffusion Hypothesis*

**Slow Diffusion Guidance**

- Link the signal to gradual information incorporation and underreaction.
- Cite seminal work on limited attention / gradual diffusion (e.g., news dispersion, investor inattention).
- Predict return continuation concentrated where attention/frictions are stronger (e.g., small, low-analyst coverage).

### A.2.2 *Production-Based Asset Pricing Hypothesis*

**Production-Based Guidance**

- Map the signal to marginal costs, investment frictions, or conditional risk premia in a production economy.
- Discuss how the signal proxies for exposure to shocks to productivity, adjustment costs, or investment.
- Relate cross-sectional pricing to production-based factors.

### A.2.3 *Consumption-Based Asset Pricing Hypothesis*

**Consumption-Based Guidance**

- Tie the signal to exposure to consumption risk or marginal utility (e.g., long-run risks, habit).
- Discuss state dependence (good versus bad times) and disaster/macro sensitivity.
- Relate to consumption-based factors and intertemporal marginal rate of substitution.

## A.3 *Data Section Generation Prompt*

The data section generation prompt was designed to create academic-style descriptions of signal construction. The prompt begins with

**Data Section Prompt—Introduction**

Write a Data section for an academic finance paper that describes how we construct the signal "[signal_name]".

The prompt provides signal construction details:

**Signal Construction Details**

SIGNAL CONSTRUCTION DETAILS:

- Signal Name: [signal_name]
- Construction: [construction_desc]
- Formula: [formula_desc]
- Numerator: [numer_full] (COMPUSTAT item: [numer])
- Denominator: [denom_full] (COMPUSTAT item: [denom])

The required content is structured in three sections:

**Required Content—Part 1: Data Source**

REQUIRED CONTENT:
1. **Data Source** (1 short paragraph):
   - State that data comes from COMPUSTAT
   - Mention coverage of publicly traded US companies
   - Note the sample period (can be general, e.g., "our sample spans several decades")

**Required Content—Part 2: Variable Definitions**

2. **Variable Definitions** (1 paragraph):
   - Define [numer_full] ([numer]): What does this variable represent? What does it measure?
   - Define [denom_full] ([denom]): What does this variable represent? What does it measure?
   - Be ACCURATE–only state facts about what these COMPUSTAT items actually measure
   - Do NOT make up information about the variables

**Required Content—Part 3: Signal Construction**

3. **Signal Construction** (1 paragraph):
   - Explain the mathematical construction: [formula_desc]
   - Explain the ECONOMIC INTUITION: What does this ratio/change capture?
   - Mention frequency (e.g., "annual observations using fiscal year-end values")
   - Note any data cleaning steps (e.g., "we require non-missing values for both variables")

Writing guidelines are specified as

**Writing Guidelines**

WRITING GUIDELINES:
- Keep it concise: 3 paragraphs, approximately 300–400 words total
- Use active voice: "We construct the signal…" not "The signal is constructed…"
- Be precise: Only make claims about variables that are factually correct
- Use present tense for methodology
- Include proper COMPUSTAT variable codes in parentheses
- Match academic style from Journal of Finance or Review of Financial Studies

The prompt specifies content to avoid:

**Content to Avoid**

AVOID:
- Making up variable definitions
- Overly detailed descriptions of COMPUSTAT database structure
- Speculation about why the signal might work (save for introduction)
- References to specific time periods unless necessary

Finally, the response format is specified as

**Response Format**

Please provide the data section in JSON format inside
<response></response> XML tags:

```
{
  "data_section": "Your complete data section text here
  (3 paragraphs)."
}
```

### A.4 *Conclusion Generation Prompt*

The conclusion section was generated using the following prompt:

**Conclusion Section Prompt**

Write a conclusion for a financial research paper analyzing the signal '[signal_name]' in predicting stock returns. Summarize the key findings of the analysis, discussing the significance of the signal in terms of predictive power and practical implications. Conclude with suggestions for future research and limitations of this study. The conclusion should be based on the following abstract:
'[abstract_text]'

### A.5 *Response Format*

All prompts required structured JSON responses within XML tags to ensure consistent parsing:

**Standard Response Format**

Please provide the [section] in JSON format inside <response></response>
XML tags:

```
“‘json
{
  "field1": "content1",
  "field2": "content2",
  ...
  "bib_entries": {
    "CitationKey1": "BibTeX entry for CitationKey1",
    "CitationKey2": "BibTeX entry for CitationKey2",
    ...
  }
}
“‘
```

## *Appendix B. Variable Definitions and Generated Paper Titles*

TABLE B1
DEFINITIONS OF ACCOUNTING VARIABLES

| Var | Definition | Var | Definition |
|---|---|---|---|
| ACCHG | Accounting changes—cumulative effect | ACO | Current assets other total |
| ACOX | Current assets other sundry | ACT | Current assets—total |
| AM | Amortization of intangibles | AO | Assets—other |
| AOLOCH | Assets and liabilities other net change | AOX | Assets—other—sundry |
| AP | Accounts payable—trade | APALCH | Accounts payable and accrued liabilities increase/decrease |
| AQC | Acquisitions | AQI | Acquisitions income contribution |
| AQS | Acquisitions sales contribution | AT | Assets—total |
| BAST | Average short-term borrowing | CAPS | Capital surplus/share premium reserve |
| CAPX | Capital expenditure | CAPXV | Capital expenditure property, plant, and equipment schedule V |
| CEQ | Common/ordinary equity—total | CEQL | Common equity liquidation value |
| CEQT | Common equity tangible | CH | Cash |
| CHE | Cash and short-term investments | CHECH | Cash and cash equivalents increase/decrease |
| CLD2 | Capitalized leases—due in second year | CLD3 | Capitalized leases—due in third year |
| CLD4 | Capitalized leases—due in fourth year | CLD5 | Capitalized leases—due in fifth year |
| COGS | Cost of goods sold | CSTK | Common/ordinary stock (capital) |
| CSTKCV | Common stock-carrying value | CSTKE | Common stock equivalents—dollar savings |
| DC | Deferred charges | DCLO | Debt capitalized lease obligations |
| DCOM | Deferred compensation | DCPSTK | Convertible debt and stock |
| DCVSR | Debt senior convertible | DCVSUB | Debt subordinated convertible |
| DCVT | Debt—convertible | DD | Debt debentures |
| DD1 | Long-term debt due in one year | DD2 | Debt due in second year |
| DD3 | Debt due in third year | DD4 | Debt due in fourth year |
| DD5 | Debt due in fifth year | DFS | Debt finance subsidiary |
| DFXA | Depreciation of tangible fixed assets | DILADJ | Dilution adjustment |
| DILAVX | Dilution available excluding extraordinary items | DLC | Debt in current liabilities—total |
| DLCCH | Current debt changes | DLTIS | Long-term debt issuance |
| DLTO | Other long-term debt | DLTP | Long-term debt tied to prime |
| DLTR | Long-term debt reduction | DLTT | Long-term debt—total |
| DM | Debt mortgages and other secured | DN | Debt notes |
| DO | Income (loss) from discontinued operations | DONR | Nonrecurring discontinued operations |
| DP | Depreciation and amortization | DPACT | Depreciation, depletion, and amortization |
| DPC | Depreciation and amortization (cash flow) | DPVIEB | Depreciation ending balance (schedule VI) |
| DPVIO | Depreciation other changes (schedule VI) | DPVIR | Depreciation retirements (schedule VI) |
| DRC | Deferred revenue current | DS | Debt—subordinated |
| DUDD | Debt unamortized debt discount and other | DV | Cash dividends (cash flow) |
| DVC | Dividends common/ordinary | DVP | Dividends—preferred/preference |
| DVPA | Preferred dividends in arrears | DVPIBB | Depreciation beginning balance (schedule VI) |
| DVT | Dividends—total | DXD2 | Debt (excl capitalized leases) due in second year |
| DXD3 | Debt (excl capitalized leases) due in third year | DXD4 | Debt (excl capitalized leases) due in fourth year |
| DXD5 | Debt (excl capitalized leases) due in fifth year | EBIT | Earnings before interest and taxes |
| EBITDA | Earnings before interest | ESOPCT | Employee stock ownership plan (ESOP) obligation (common)—total |

(continued)

TABLE B1 (continued)
Definitions of Accounting Variables

| Var | Definition | Var | Definition |
|---|---|---|---|
| ESOPDLT | ESOP debt—long term | ESOPT | Preferred ESOP obligation—total |
| ESUB | Equity in earnings—unconsolidated subsidiaries | ESUBC | Equity in net loss earnings |
| EXRE | Exchange rate effect | FATB | Property, plant, and equipment buildings |
| FATC | Property, plant, and equipment construction in progress | FATE | Property, plant, equipment and machinery equipment |
| FATL | Property, plant, and equipment leases | FATN | Property, plant, equipment, and natural resources |
| FATO | Property, plant, and equipment other | FATP | Property, plant, equipment, and land improvements |
| FIAO | Financing activities other | FINCF | Financing activities net cash flow |
| FOPO | Funds from operations other | FOPOX | Funds from operations—other excl option tax benefit |
| FOPT | Funds from operations total | FSRCO | Sources of funds other |
| FSRCT | Sources of funds total | FUSEO | Uses of funds other |
| FUSET | Uses of funds total | GDWL | Goodwill |
| GP | Gross profit (loss) | IB | Income before extraordinary items (IB) |
| IBADJ | IB adjusted for common stock equivalents | IBC | Income before extraordinary items (cash flow) |
| IBCOM | Income before extraordinary items available for common | ICAPT | Invested capital—total |
| IDIT | Interest and related income—total | INTAN | Intangible assets—total |
| INTC | Interest capitalized | INTPN | Interest paid net |
| INVCH | Inventory decrease (increase) | INVFG | Inventories finished goods |
| INVO | Inventories other | INVRM | Inventories raw materials |
| INVT | Inventories—total | INVWIP | Inventories work in progress |
| ITCB | Investment tax credit (balance sheet) | ITCI | Investment tax credit (income account) |
| IVACO | Investing activities other | IVAEQ | Investment and advances—equity |
| IVAO | Investment and advances other | IVCH | Increase in investments |
| IVNCF | Investing activities net cash flow | IVST | Short-term investments—total |
| IVSTCH | Short-term investments change | LCO | Current liabilities—other—total |
| LCOX | Current liabilities other sundry | LCOXDR | Current liabilities—other—excl deferred revenue |
| LCT | Current liabilities—total | LIFR | LIFO reserve |
| LO | Liabilities—other—total | LT | Liabilities—total |
| MIB | Minority interest (balance sheet) | MII | Minority interest (income account) |
| MRC1 | Rental commitments minimum first year | MRC2 | Rental commitments minimum second year |
| MRC3 | Rental commitments minimum third year | MRC4 | Rental commitments minimum fourth year |
| MRC5 | Rental commitments minimum fifth year | MRCT | Rental commitments minimum 5-year total |
| MSA | Marketable securities adjustment | NI | Net income (loss) |
| NIADJ | Net income adjusted for common stock equiv. | NIECI | Net income effect capitalized interest |
| NOPI | Nonoperating income (expense) | NOPIO | Nonoperating income (expense) other |
| NP | Notes payable short-term borrowings | OANCF | Operating activities net cash flow |
| OB | Order backlog | OIADP | Operating income after depreciation |
| PI | Pretax income | PIDOM | Pretax income domestic |
| PIFO | Pretax income foreign | PPEGT | Property, plant, and equipment—total (gross) |
| PPENB | Property, plant, and equipment buildings (net) | PPENC | Property, plant, equipment, construction in progress (net) |
| PPENLI | Property, plant, equipment, land, and improvements (net) | PPENME | Property, plant, equipment, machinery, and equipment (net) |

(continued)

TABLE B1 (continued)
DEFINITIONS OF ACCOUNTING VARIABLES

| Var | Definition | Var | Definition |
|---|---|---|---|
| PPENNR | Property, plant, equipment, natural resources (net) | PPENO | Property, plant, and equipment, other (net) |
| PPENT | Property, plant, and equipment—total (net) | PPEVBB | Property, plant, equipment, beginning balance (schedule V) |
| PPEVEB | Property, plant, and equipment, ending balance | PPEVO | Property, plant, and equipment, other changes (schedule V) |
| PPEVR | Property, plant, and equipment, retirements (schedule V) | PRSTKC | Purchase of common and preferred stock |
| PSTK | Preferred/preference stock (capital)—total | PSTKC | Preferred stock convertible |
| PSTKL | Preferred stock liquidating value | PSTKN | Preferred/preference stock—nonredeemable |
| PSTKR | Preferred/preference stock—redeemable | PSTKRV | Preferred stock redemption value |
| RDIP | In-process research and development expense | RE | Retained earnings |
| REA | Retained earnings, restatement | REAJO | Retained earnings, other adjustments |
| RECCH | Accounts receivable decrease (increase) | RECCO | Receivables—current—other |
| RECD | Receivables—estimated doubtful | RECT | Receivables—total |
| RECTA | Retained earnings, cumulative translation adjustment | RECTR | Receivables—trade |
| REUNA | Retained earnings, unadjusted | SALE | Sales/turnover (net) |
| SEQ | Stockholders' equity—total | SIV | Sale of investments |
| SPI | Special items | SPPE | Sale of property |
| SPPIV | Sale of property, plant, equipment, investments gain (loss) | SSTK | Sale of common and preferred stock |
| TLCF | Tax loss carry forward | TSTK | Treasury stock—total (all capital) |
| TSTKC | Treasury stock—common | TSTKP | Treasury stock—preferred |
| TXACH | Income taxes accrued increase/decrease | TXBCO | Excess tax benefit stock options—cash flow |
| TXC | Income tax—current | TXDB | Deferred taxes (balance sheet) |
| TXDBA | Deferred tax asset—long term | TXDBCA | Deferred tax asset—current |
| TXDBCL | Deferred tax liability—current | TXDC | Deferred taxes (cash flow) |
| TXDFED | Deferred taxes—federal | TXDFO | Deferred taxes—foreign |
| TXDI | Income tax—deferred | TXDITC | Deferred taxes and investment tax credit |
| TXDS | Deferred taxes—state | TXFED | Income tax federal |
| TXFO | Income tax foreign | TXNDB | Net deferred tax asset (liab.)—total |
| TXNDBA | Net deferred tax asset | TXNDBL | Net deferred tax liability |
| TXNDBR | Deferred tax residual | TXO | Income taxes—other |
| TXP | Income tax payable | TXPD | Income taxes paid |
| TXR | Income tax refund | TXS | Income tax state |
| TXT | Income tax total | TXW | Excise taxes |
| WCAP | Working capital (balance sheet) | WCAPC | Working capital change, other increase/decrease |
| WCAPCH | Working capital change, total | XACC | Accrued expenses |
| XAD | Advertising expense | XDEPL | Depletion expense (schedule VI) |
| XI | Extraordinary items | XIDO | Ext. items and discontinued operations |
| XIDOC | Ext. items and disc. operations (cash flow) | XINT | Interest and related expenses—total |
| XOPR | Operating expenses—total | XPP | Prepaid expenses |
| XPR | Pension and retirement expense | XRD | Research and development expense |
| XRENT | Rental expense | XSGA | Selling, general and administrative expense |
| EMP | Number of Employees | ME | Market equity at fiscal year end |

TABLE B2
Signal Descriptions and Paper Titles–Part 1

| Numerator | Denominator | Signal | Title | Acronym |
|---|---|---|---|---|
| ACO | SEQ | ratio | Liquidity Leverage Intensity and the Cross Section of Stock Returns | LLI |
| ACOX | CEQL | ratio | Sundry Asset Coverage and the Cross Section of Stock Returns | SAC |
| ACOX | ICAPT | ratio | Miscellaneous Capital Intensity and the Cross Section of Stock Returns | MCI |
| ACOX | SEQ | ratio | Residual Asset Burden and the Cross Section of Stock Returns | RAB |
| AOLOCH | DPACT | ratio | Accrual Depreciation Sensitivity and the Cross Section of Stock Returns | ADS |
| AOLOCH | LT | ratio | Balance Sheet Velocity and the Cross Section of Stock Returns | BSV |
| AOLOCH | XINT | ratio | Debt Service Flexibility and the Cross Section of Stock Returns | DSF |
| AQC | IVAO | diff | Acquisition Investment Discipline and the Cross Section of Stock Returns | AID |
| AQC | WCAP | diff | Acquisition Capacity Utilization and the Cross Section of Stock Returns | ACU |
| CAPS | XSGA | ratio | Operational Funding Efficiency and the Cross Section of Stock Returns | OFE |
| CAPX | NOPIO | diff | Nonoperating Investment Sensitivity and the Cross Section of Stock Returns | NIS |
| CAPXV | NOPI | diff | Capex Nonoperating Dependence and the Cross Section of Stock Returns | CND |
| CEQ | COGS | ratio | Equity Production Intensity and the Cross Section of Stock Returns | EPI |
| CEQL | CHE | diff | Equity Liquidity Absorption and the Cross Section of Stock Returns | ELA |
| CH | EBIT | ratio | Cash Hoarding Intensity and the Cross Section of Stock Returns | CHI |
| CH | EBITDA | ratio | Cash Conversion Cushion and the Cross Section of Stock Returns | CCC |
| CH | OIADP | ratio | Operating Liquidity Coverage and the Cross Section of Stock Returns | OLC |
| COGS | CSTK | ratio | Equity Operational Burden and the Cross Section of Stock Returns | EOB |
| CSTK | ACT | diff | Equity Dilution Pressure and the Cross Section of Stock Returns | EDP |
| CSTK | AO | diff | Equity Financing Intensity and the Cross Section of Stock Returns | EFI |
| CSTK | AOX | diff | Equity Issuance Restraint and the Cross Section of Stock Returns | EIR |
| CSTK | AT | diff | Equity Capital Discipline and the Cross Section of Stock Returns | ECD |
| CSTK | CAPS | diff | Capital Reserve Depletion and the Cross Section of Stock Returns | CRD |
| CSTK | CEQ | diff | Share Base Stability and the Cross Section of Stock Returns | SBS |
| CSTK | DLC | diff | Equity Leverage Sensitivity and the Cross Section of Stock Returns | ELS |
| CSTK | DP | diff | Equity Depreciation Absorption and the Cross Section of Stock Returns | EDA |
| CSTK | DPACT | diff | Equity Preservation Intensity and the Cross Section of Stock Returns | EPI |
| CSTK | DVC | diff | Dividend Coverage Erosion and the Cross Section of Stock Returns | DCE |
| CSTK | DVT | diff | Payout Retention Signal and the Cross Section of Stock Returns | PRS |
| CSTK | EMP | diff | Employee Dilution Control and the Cross Section of Stock Returns | EDC |
| CSTK | GP | diff | Profit Dilution Resistance and the Cross Section of Stock Returns | PDR |
| CSTK | ICAPT | diff | Equity Financing Restraint and the Cross Section of Stock Returns | EFR |

TABLE B2
Signal Descriptions and Paper Titles–Part 2

| Numerator | Denominator | Signal | Title | Acronym |
|---|---|---|---|---|
| CSTK | LT | diff | Equity Dilution Burden and the Cross Section of Stock Returns | EDB |
| CSTK | PPEGT | diff | Capital Structure Discipline and the Cross Section of Stock Returns | CSD |
| CSTK | SALE | diff | Revenue Dilution Avoidance and the Cross Section of Stock Returns | RDA |
| CSTK | SALE | ratio | Revenue Equity Efficiency and the Cross Section of Stock Returns | REE |
| CSTK | XSGA | diff | Operational Equity Absorption and the Cross Section of Stock Returns | OEA |
| CSTK | ME | diff | Share Issuance Restraint and the Cross Section of Stock Returns | SIR |
| DLC | LT | diff | Maturity Structure Stability and the Cross Section of Stock Returns | MSS |
| DLTIS | ACT | diff | Liquidity Leverage Burden and the Cross Section of Stock Returns | LLB |
| DLTIS | AT | diff | Debt Expansion Restraint and the Cross Section of Stock Returns | DER |
| DLTIS | CAPS | diff | Equity Buffer Preservation and the Cross Section of Stock Returns | EBP |
| DLTIS | CAPX | diff | Investment Financing Conservatism and the Cross Section of Stock Returns | IFC |
| DLTIS | CAPXV | diff | Capital Intensity Alignment and the Cross Section of Stock Returns | CIA |
| DLTIS | CEQ | diff | Leverage Growth Discipline and the Cross Section of Stock Returns | LGD |
| DLTIS | CEQL | diff | Debt Capacity Utilization and the Cross Section of Stock Returns | DCU |
| DLTIS | DP | diff | Depreciation Financing Prudence and the Cross Section of Stock Returns | DFP |
| DLTIS | DPACT | diff | Depreciation Coverage Moderation and the Cross Section of Stock Returns | DCM |
| DLTIS | EBIT | diff | Earnings Debt Prudence and the Cross Section of Stock Returns | EDP |
| DLTIS | EBITDA | diff | Earnings Leverage Stability and the Cross Section of Stock Returns | ELS |
| DLTIS | GP | diff | Profit Borrowing Restraint and the Cross Section of Stock Returns | PBR |
| DLTIS | ICAPT | diff | Leverage Expansion Intensity and the Cross Section of Stock Returns | LEI |
| DLTIS | PPEGT | diff | Asset Financing Restraint and the Cross Section of Stock Returns | AFR |
| DLTIS | PPENT | diff | Fixed Asset Conservatism and the Cross Section of Stock Returns | FAC |
| DLTIS | SALE | diff | Revenue Debt Discipline and the Cross Section of Stock Returns | RDD |
| DLTIS | SEQ | diff | Equity Dilution Protection and the Cross Section of Stock Returns | EDP |
| DLTIS | XOPR | diff | Operating Leverage Discipline and the Cross Section of Stock Returns | OLD |
| DLTIS | XRENT | diff | Lease Financing Moderation and the Cross Section of Stock Returns | LFM |
| FINCF | CSTK | diff | Financing Activity Intensity and the Cross Section of Stock Returns | FAI |
| FINCF | PPEGT | diff | Capital Funding Prudence and the Cross Section of Stock Returns | CFP |
| ICAPT | NOPIO | diff | Capital Expansion Intensity and the Cross Section of Stock Returns | CEI |
| ICAPT | XSGA | ratio | Overhead Absorption Efficiency and the Cross Section of Stock Returns | OAE |

TABLE B2
Signal Descriptions and Paper Titles–Part 3

| Numerator | Denominator | Signal | Title | Acronym |
|---|---|---|---|---|
| INVT | NP | diff | Inventory Financing Burden and the Cross Section of Stock Returns | IFB |
| INVT | XSGA | diff | Inventory Overhead Efficiency and the Cross Section of Stock Returns | IOE |
| ITCI | ME | ratio | Tax Subsidy Intensity and the Cross Section of Stock Returns | TSI |
| IVCH | INTAN | diff | Intangible Investment Restraint and the Cross Section of Stock Returns | IIR |
| NP | CEQT | diff | Debt Capacity Preservation and the Cross Section of Stock Returns | DCP |
| OANCF | CEQ | ratio | Equity Cash Generation and the Cross Section of Stock Returns | ECG |
| OANCF | CEQL | ratio | Equity Yield Rate and the Cross Section of Stock Returns | EYR |
| OANCF | CSTK | ratio | Shareholder Cash Productivity and the Cross Section of Stock Returns | SCP |
| OANCF | DLC | ratio | Debt Service Coverage and the Cross Section of Stock Returns | DSC |
| OANCF | DPACT | ratio | Cash Generation Multiple and the Cross Section of Stock Returns | CGM |
| OANCF | PPEGT | diff | Asset Cash Intensity and the Cross Section of Stock Returns | ACI |
| OANCF | SEQ | ratio | Equity Cash Return and the Cross Section of Stock Returns | ECR |
| OIADP | DVT | diff | Payout Earnings Sensitivity and the Cross Section of Stock Returns | PES |
| PPENT | DP | diff | Capital Replacement Intensity and the Cross Section of Stock Returns | CRI |
| PPENT | NOPIO | diff | Capital Nonoperating Sensitivity and the Cross Section of Stock Returns | CNS |
| PPENT | XOPR | diff | Capital Expansion Burden and the Cross Section of Stock Returns | CEB |
| PPENT | XRENT | diff | Asset Lease Intensity and the Cross Section of Stock Returns | ALI |
| PRSTKC | CSTK | ratio | Equity Buyback Intensity and the Cross Section of Stock Returns | EBI |
| PRSTKC | DVC | ratio | Buyback Dividend Preference and the Cross Section of Stock Returns | BDP |
| PRSTKC | DVT | ratio | Repurchase Distribution Tilt and the Cross Section of Stock Returns | RDT |
| PRSTKC | NOPI | ratio | Nonoperating Buyback Reliance and the Cross Section of Stock Returns | NBR |
| PRSTKC | XINT | ratio | Buyback Debt Coverage and the Cross Section of Stock Returns | BDC |
| RECD | NOPIO | diff | Collection Risk Absorption and the Cross Section of Stock Returns | CRA |
| SALE | TXT | diff | Revenue Tax Sensitivity and the Cross Section of Stock Returns | RTS |
| SEQ | CHE | diff | Equity Dilution Intensity and the Cross Section of Stock Returns | EDI |
| TXC | DVC | ratio | Tax Payout Burden and the Cross Section of Stock Returns | TPB |
| TXDFED | EBIT | ratio | Tax Deferral Burden and the Cross Section of Stock Returns | TDB |
| TXDFED | OIADP | ratio | Tax Shield Intensity and the Cross Section of Stock Returns | TSI |
| TXFED | DVC | ratio | Tax Distribution Pressure and the Cross Section of Stock Returns | TDP |
| TXFED | DVT | ratio | Dividend Tax Friction and the Cross Section of Stock Returns | DTF |
| TXFED | ME | ratio | Tax Equity Burden and the Cross Section of Stock Returns | TEB |
| XINT | CEQT | diff | Interest Burden Relief and the Cross Section of Stock Returns | IBR |
| XOPR | TXT | diff | Expense Tax Elasticity and the Cross Section of Stock Returns | ETE |

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